United States Patent Application Publication	20250281731
Kind Code	A1
Publication Date	September 11, 2025
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UNIVERSAL SINGLE-USE CAP FOR MALE AND FEMALE CONNECTORS

Abstract

A device for connection to a medical connector, the device includes a cap having an annular wall having an exterior wall surface and an interior wall surface including a plurality of inwardly extending protrusions sized and shaped to frictionally engage a male luer, and a peelable seal. The cap configured to define a chamber to contain an absorbent material and disinfectant or antimicrobial agent. The cap includes one or more threads adapted to engage with a female luer connector. The cap is adapted to engage a male luer connector in a press-fit connection. The peelable seal prevents the disinfectant or the antimicrobial agent from exiting the chamber. Also described are methods of disinfecting a medical connector, and apparatuses comprising a device for disinfecting a medical connector.

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Appl. No.: **19/216867**

Filed: **May 23, 2025**

Related U.S. Application Data

parent US continuation 17045840 20201007 parent-grant-document US 12329924 US continuation PCT/US2019/026536 20190409 child US 19216867
us-provisional-application US 62655512 20180410

Publication Classification

Int. Cl.: **A61M39/16** (20060101); **A61M39/20** (20060101)

U.S. Cl.:

CPC **A61M39/162** (20130101); **A61M39/20** (20130101); A61M2205/02 (20130101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. application Ser. No. 17/045,840 filed on Oct. 7, 2020, now allowed, which is the national stage entry of PCT/US2019/026536, filed on Apr. 9, 2019, which claims priority from provisional U.S. Patent Application No. 62/655,512 filed Apr. 10, 2018, the disclosures of which are hereby incorporated by reference in their entirety.

TECHNICAL FIELD
[0002] The present disclosure generally relates to a device for disinfecting and sterilizing access ports with, e.g., male and female luer fitting, and, in particular, to disinfecting and sterilizing devices capable of accommodating multiple types of connectors.
BACKGROUND
[0003] Vascular access devices (VAD's) are commonly used therapeutic devices and include intravenous (IV) catheters. There are two general classifications of VAD's, peripheral catheters and central venous catheters. Bacteria and other microorganisms may gain entry into a patient's vascular system from access hubs and ports/valves upon connection to the VAD to deliver the fluid or pharmaceutical. Each access hub (or port/valve or connection) is associated with some risk of transmitting a catheter related bloodstream infection (CRBSI), which can be costly and potentially lethal.
[0004] In order to decrease CRBSI cases and to ensure VAD's are used and maintained correctly, standards of practice have been developed, which include disinfecting and cleaning procedures.
[0005] Disinfection caps have been added to the Society for Healthcare Epidemiology of America (SHEA) guidelines and early indications are that caps will also be incorporated into the 2016 Infusion Nurses Standards (INS) guidelines.
[0006] In developed markets, when utilizing an IV catheter, a needleless connector will typically be used to close off the system and then subsequently accessed to administer medication or other necessary fluids via the catheter to the patient. INS Standards of Practice recommend the use of a needleless connector and state that it should be “consistently and thoroughly disinfected using alcohol, tincture of iodine or chlorhexidine gluconate/alcohol combination prior to each access.” The disinfection of the needleless connector is ultimately intended to aid in the reduction of bacteria that could be living on the surface and possibly lead to a variety of catheter related complications including CRBSI. Nurses will typically utilize a 70% isopropyl alcohol (IPA) pad to complete this disinfection task by doing what is known as “scrubbing the hub.” However, compliance to this practice is typically very low. In addition to a lack of compliance to “scrubbing the hub”, it has also been noted through clinician interviews that there is often a variation in scrub time, dry time and the number of times the needleless connector is scrubbed.
[0007] Throughout the sequence of procedures associated with the transmission of a microorganism that can cause a CRBSI, there are many risks of contact or contamination. Contamination can occur during drug mixing, attachment of a cannula, and insertion into the access hub. Because the procedure to connect to a VAD is so common and simple, the risk associated with entry into a patient's vascular system has often been overlooked. Presently, the risk to hospitals and patients is a substantial function of the diligence of the clinician performing the connection, and

this diligence is largely uncontrollable.

[0008] Currently, caps for male needleless connectors, female needleless connectors, intravenous (IV), and hemodialysis lines use different designs and are therefore limited to the types of connectors to which the cap can be attached. Thus, prior disinfecting caps were designed to fit one type of connector only, and were specific to one particular size and/or shape of connector. Thus, there is a need for a disinfecting device capable of accommodating multiple types of connectors to streamline the disinfecting process. There is also a need for a disinfecting device capable of continuous disinfection for multiple days.

SUMMARY

[0009] One aspect of the present disclosure pertains to device for connection to a medical connector according to an exemplary embodiment of the present disclosure generally comprises a cap, absorbent material, a disinfectant or the antimicrobial agent and a peelable seal. The cap comprises an integral body, a closed end, an annular wall having a length extending from the closed end to an open end that defines a chamber containing an absorbent material and disinfectant or antimicrobial agent. The open end defines an end face and an engagement surface.

[0010] The annular wall of the cap comprises an exterior wall surface and an interior wall surface. The interior wall surface comprises a plurality of inwardly extending protrusions sized and shaped to frictionally engage a male luer connector. The interior wall surface also comprises internal threads adjacent the closed end. The internal threads are adapted and sized to engage a female luer connector.

[0011] In one or more embodiments, the plurality of inwardly extending protrusions are sized to define an opening having a diameter that is dilatable from an initial diameter of from about 7-9 mm to a dilated diameter of about 9-12 mm to accept a collar of a male connectors. In one or more embodiments, the male luer connector frictionally engages the inwardly extending protrusions opening via a press-fit connection upon insertion into the chamber.

[0012] In one or more embodiments, the plurality of inwardly extending protrusions extends along an entire length of the interior wall surface of the cap. In yet another embodiment, the plurality of inwardly extending partially extends along the length of the interior wall surface of the cap.

[0013] In one or more embodiments, the plurality of inwardly extending protrusions is elongate. In one or more embodiments, the plurality of inwardly extending protrusions is tapered.

[0014] In one or more embodiments, the opening adjacent the open end of the interior wall surface is sized and adapted to receive a male luer connector in a press-fit connection.

[0015] The interior wall surface comprises internal threads adjacent to the closed end. The internal threads are adapted and sized to engage a female luer connector. In one or more embodiments, the internal threads adjacent the closed end of the cap partially extend along a length of the interior wall surface of the cap.

[0016] The absorbent material and the disinfectant or the antimicrobial agent contacts the male luer connector, the female luer connector and the hemodialysis connector after insertion of the connector through the dilatable opening.

[0017] The peelable seal is disposed on the end face of the cap to prevent the disinfectant or the antimicrobial agent from exiting the chamber.

[0018] In one or more embodiments, the female luer connector is selected from the group consisting of needle-free connectors, stopcocks, and hemodialysis connectors.

[0019] In one or more embodiments, the male connector is an intravenous tubing end or stopcock.

[0020] In one or more embodiments, the male luer connector rests on the peripheral ledge upon being fully inserted in into the chamber.

[0021] In one or more embodiments, the chamber comprises a first portion adjacent the closed end having a first portion diameter and a second portion adjacent the open end having a second portion diameter, the first portion and second portion in fluid communication with each other and the first portion diameter being less than the second portion diameter. In one or more embodiments, the

annular wall of the cap is frusto-conically shaped.
[0022] In one or more embodiments, the cap comprises a polypropylene or polyethylene material. In one or more embodiments, the exterior cap surface includes a plurality of grip members.
[0023] In one or more embodiments, the absorbent material is under radial compression by the internal threads to retain the absorbent material in the chamber. In one or more embodiments, the absorbent material is retained within the chamber without being radially compressed from the internal threads. In one or more embodiments, the absorbent material is a nonwoven material, foam or a sponge. In a specific embodiment, the foam is a polyurethane foam.
[0024] In one or more embodiments, the cap comprises a polypropylene or polyethylene material.
[0025] In one or more embodiments, the disinfectant or antimicrobial agent is selected from the group consisting of isopropyl alcohol, ethanol, **2**-propanol, butanol, methylparaben, ethylparaben, propylparaben, propyl gallate, butylated hydroxyanisole (BHA), butylated hydroxytoluene, t-butyl-hydroquinone, chloroxylenol, chlorohexidine, chlorhexidine diacetate, chlorohexidine gluconate, povidone iodine, alcohol, dichlorobenzyl alcohol, dehydroacetic acid, hexetidine, triclosan, hydrogen peroxide, colloidal silver, benzethonium chloride, benzalkonium chloride, octenidine, antibiotic and mixtures thereof. In a specific embodiment, the disinfectant or antimicrobial agent comprises at least one of chlorhexidine gluconate and chlorhexidine diacetate. In one or more embodiments, the disinfectant or antimicrobial agent is a fluid or a gel.
[0026] Compression of the absorbent material toward the closed end of the chamber upon connection to the female luer connector or the male luer connector allows the connector to contact the disinfectant or antimicrobial agent to disinfect the female luer connector or the male luer connector. When a male luer connector, a female luer connector or hemodialysis connector is inserted through the open end, the absorbent material and the disinfectant or the antimicrobial agent contact the male luer connector, female luer connector or hemodialysis connector.
[0027] In one or more embodiments, the cap further comprises a peripheral ledge extending radially inward from the annular wall.
[0028] In one or more embodiments, the peelable seal comprises an aluminum or multi-layer polymer film peel back top. In a specific embodiment, the peelable seal is heat-sealed or induction sealed to the cap open end.
[0029] A second aspect of the present disclosure pertains to methods of disinfecting medical connectors. In one or more embodiments, a method of disinfecting a medical connector comprises connecting the device of one or more embodiments to a medical connector, wherein connecting includes frictionally engaging the interior wall surface upon insertion into the chamber or engaging the internal threads, such that the medical connector contacts the absorbent material and the disinfectant or antimicrobial agent.
[0030] A third aspect of the present disclosure pertains to an assembly. The assembly comprises the device of one or more embodiments connected to a medical connector. In one or more embodiments, the medical connector is selected from a male luer connector, a female luer connector, and needleless connector.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0031] FIG. **1** shows a top perspective view of an embodiment of a device of the present disclosure;
[0032] FIG. **2** shows a top perspective elevation view of an embodiment of a device of the present disclosure;
[0033] FIG. **3** shows a partial sectional view of a device according to an embodiment of the present disclosure;

[0034] FIG. **4** shows a partial sectional view showing connection of the device of FIG. **1** to a male connector;
[0035] FIG. **5** shows a partial sectional view showing connection of the device of FIG. **1** to a female Luer stopcock;
[0036] FIG. **6** shows a partial sectional view showing connection of the device of FIG. **1** to a female Luer MaxZero® connector;
[0037] FIG. **7** shows a partial sectional view showing connection of the device of FIG. **1** to a Q-Syte® connector;
[0038] FIG. **8** shows a top perspective view of an embodiment of a device of the present disclosure;
[0039] FIG. **9** shows a top perspective elevation view of an embodiment of a device of the present disclosure;
[0040] FIG. **10** shows a partial sectional view of a device according to an embodiment of the present disclosure;
[0041] FIG. **11** shows a partial sectional view showing connection of the device of FIG. **8** to a male connector;
[0042] FIG. **12** shows a partial sectional view showing connection of the device of FIG. **8** to a female Luer stopcock;
[0043] FIG. **13** shows a partial sectional view showing connection of the device of FIG. **8** to a female Luer MaxZero;
[0044] FIG. **14** shows a partial sectional view showing connection of the device of FIG. **8**
[0045] to a Q-Syte connector;
[0046] FIG. **15** shows a perspective view of a female luer connector with septum according to the prior art;
[0047] FIG. **16** shows a perspective view a female luer connector with stopcock according to the prior art;
[0048] FIG. **17** shows a perspective view of a male luer connector according to the prior art;
[0049] FIG. **18** shows a perspective view of a hemodialysis connector according to the prior art.

DETAILED DESCRIPTION

[0050] Before describing several exemplary embodiments of the disclosure, it is to be understood that the disclosure is not limited to the details of construction or process steps set forth in the following description. The disclosure is capable of other embodiments and of being practiced or being carried out in various ways.
[0051] With respect to terms used in this disclosure, the following definitions are provided.
[0052] As used herein, the use of “a,” “an,” and “the” includes the singular and plural.
[0053] As used herein, the term “catheter related bloodstream infection” or “CRBSI” refers to any infection resulting from the presence of a catheter or IV line.
[0054] As used herein, the term “Luer connector” refers to a connection collar that is the standard way of attaching syringes, catheters, hubbed needles, IV tubes, etc. to each other. The Luer connector consists of male and female interlocking tubes, slightly tapered to hold together better with even just a simple pressure/twist fit. Luer connectors can optionally include an additional outer rim of threading, allowing them to be more secure. The Luer connector male end is generally associated with a flush syringe and can interlock and connect to the female end located on the vascular access device (VAD). A Luer connector comprises a distal end, a proximal end, an irregularly shaped outer wall, a profiled center passageway for fluid communication from the chamber of the barrel of a syringe to the hub of a VAD. A Luer connector also has a distal end channel that releasably attaches the Luer connector to the hub of a VAD, and a proximal end channel that releasably attaches the Luer connector to the barrel of a syringe.
[0055] Embodiments of the disclosure pertain to a universal single-use device for connection to and disinfection of a medical connector, including male luer connectors and female luer connectors, in

which the device comprises a cap, absorbent material, a disinfectant or the antimicrobial agent and a peelable seal. Embodiments of the disclosure pertain to a device that provides a mechanical barrier for connectors and contains an antimicrobial agent for disinfection. Embodiments of the disclosure pertain to a device that allows a practitioner to streamline the disinfecting process.

[0056] The assembled device **100** according to one embodiment is shown in FIGS. **1-3**. FIGS. **4-7** show the device engaged with medical connectors according to embodiments of the present disclosure. The assembled device **200** according to a second embodiment is shown in FIGS. **8-10**. FIGS. **11-14** show the device of the second embodiment engaged with medical connectors according to embodiments of the present disclosure. FIGS. **15-18** show various medical connectors according to the prior art. Referring to FIGS. **1-3**, device **100** for connection to a medical connector according to an exemplary embodiment of the present disclosure generally comprises a cap **102**, absorbent material **114**, a disinfectant or an antimicrobial agent, and a peelable seal **128**. The cap **102** comprises an integral body **104**, a closed end **106**, an annular wall **108** having a length LB extending from the closed end **106** to an open end **110** that defines a chamber **112** containing an absorbent material **114** and disinfectant or antimicrobial agent. The open end **110** defines an end face **116** and an engagement surface **118**.

[0057] Referring to FIG. **3**, the annular wall **108** of the cap **102** comprises an exterior wall surface **120** and an interior wall surface **122**. The interior wall surface **122** comprises a plurality of inwardly extending protrusions **124** sized and shaped to frictionally engage a male luer connector or a female luer connector with a compatible diameter. The interior wall surface **122** also comprises internal threads **126** adjacent the closed end **106**. The internal threads **126** are adapted and sized to engage a female luer connector. The interior wall surface **122** defines an opening **150** adjacent the open end.

[0058] In one or more embodiments, the plurality of inwardly extending protrusions **124** is sized to define an opening **130** having a diameter that is dilatable. The dilatable opening **130** can be sized to frictionally engage a male luer connector or a female luer connector with a compatible diameter. In one or more embodiments, the dilatable opening **130** has a diameter that is dilatable from an initial diameter of from about 7-9 mm to a dilated diameter of about 9-12 mm to accept a collar of a male connector.

[0059] The peelable seal **128** is disposed on the end face **116** of the cap **102** to prevent the disinfectant or the antimicrobial agent from exiting the chamber **112**.

[0060] Referring to FIGS. **1** and **2**, in one or more embodiments, the exterior wall surface **122** of the cap **102** comprises a plurality of grip members **162**.

[0061] The cap **102** comprises an annular cap wall **108** having a cap wall length LB extending from a cap closed end **106** to a cap open end **110**. The cap **102** comprises an interior cap surface **164** and an exterior cap surface **166**.

[0062] The peelable seal **128** is disposed on the cap open end **110** to prevent the disinfectant or the antimicrobial agent from exiting the chamber.

[0063] Referring to FIG. **4**, in one or more embodiments, the male luer connector **140** frictionally engages the inwardly extending protrusions **124** via a press-fit connection upon insertion into the chamber **112**. To limit inadvertent movement of the male luer connector **140**, the plurality of inwardly extending protrusions **124** interferingly engage the male luer connector **140** in limiting movement thereof. With sufficient force being applied to the plurality of inwardly extending protrusions **124** by a male luer connector **140** when the male luer connector **140** is inserted into the open end **110** of the cap **102**, the plurality of inwardly extending protrusions **124** may be caused to deform or be displaced around the male luer connector **140** so as to nest the male luer connector **140** and thereby restrict movement of the male luer connector **140**. With sufficient force being applied to the plurality of inwardly extending protrusions **124** by a male luer connector **140** when the male luer connector **140** is inserted into the open end **110** of the cap **102**, the plurality of inwardly extending protrusions **124** may be deformed to envelope the male luer connector **140** so as to nest the male luer connector **140** and thereby restrict movement of the male luer connector **140**. In one or more embodiments, the plurality of inwardly extending protrusions **124** comprises 1-100 inwardly extending protrusions. In a specific embodiment, the plurality of inwardly extending protrusions **124** comprises 1-20 inwardly extending protrusions. In one or more embodiments, the plurality of inwardly extending protrusions **124** extends along an entire length of the interior wall surface **122** of the cap **102**. In yet another embodiment, the plurality of inwardly extending protrusions **124** partially extends along the length of the interior wall surface **122** of the cap **102**.

[0064] In one or more embodiments, the plurality of inwardly extending protrusions **124** is elongated. In one or more embodiments, the plurality of inwardly extending protrusions **124** is tapered.

[0065] In one or more embodiments, the cap **102** further includes a peripheral ledge **160** extending radially inward from the annular wall **108** which the male connector contacts when the male luer connector is inserted into the chamber **112**.

[0066] In one or more embodiments, the opening adjacent to the open end **110** of the interior wall surface **122** is sized and adapted to receive a male luer connector in a press-fit connection.

[0067] Referring to FIGS. **4-6**, the interior wall surface **122** comprises internal threads **126** adjacent to the closed end **106**. The internal threads **126** are adapted and sized to engage a female luer connector. In one or more embodiments, the internal threads **126** adjacent the closed end **106** of the cap **102** partially extend along a length of the interior wall surface **122** of the cap **102**.

[0068] The absorbent material **114** and the disinfectant or the antimicrobial agent contacts the male luer connector **140**, the female luer connector **142**, and the hemodialysis connector **144** after insertion of the connector through the dilatable opening **130**.

[0069] Referring to FIGS. **5** and **6**, the interior wall surface **122** comprises internal threads **126** adjacent to the closed end **106**. The internal threads **126** are adapted and sized to engage a female luer connector **142**. The absorbent material **114** and the disinfectant or the antimicrobial agent contacts the female luer connector **142** after insertion of the connector into the open end **110** of the cap **102**.

[0070] In one or more embodiments, the female connector **142** may be selected from the group consisting essentially of needle-free connectors (or needleless connectors), catheter luer connectors, stopcocks, and hemodialysis connectors. In one or more embodiments, the female luer is selected from needleless connectors/needle-free connectors. In one or more embodiments, the needleless connector is selected from a Q-Syte connector, MaxPlus, MaxPlus Clear, MaxZero, UltraSite, Caresite, InVision-Plus, Safeline, OneLink, V-Link, ClearLink, NeutraClear, Clave, MicroClave, MicroClave Clear, Neutron, NanoClave, Kendall, Nexus, InVision, Vadsite, Bionector, etc.

[0071] In one or more embodiments, the male connector **140** may be an intravenous tubing end, stopcock or male lock luer.

[0072] Referring to FIG. **4**, in one or more embodiments, the male luer connector **140** rests on the peripheral ledge **160** upon being fully inserted in into the chamber **112**.

[0073] In one or more embodiments, the internal threads **126** adjacent the closed end **106** of the cap **102** partially extend along a length of the interior wall surface **122** of the cap.

[0074] In one or more embodiments, the male luer connector **140** frictionally engages the interior wall surface **122** via a press-fit connection upon insertion into the chamber **112**.

[0075] In one or more embodiments, the opening **130** adjacent the open end **110** of the interior wall surface **122** of the cap is sized and adapted to receive a male luer connector **140** in a press-fit connection.

[0076] Referring to FIG. **7**, in one or more embodiments, the interior wall surface **122** comprises internal threads **126** adjacent to the closed end **106**. The internal threads **126** are adapted and sized to engage a female connector or needleless connector **144**. The absorbent material **114** and the disinfectant or the a ntimicrobial agent contacts the female connector or needleless connector **144** after insertion of the connector into the open end **110** of the cap **102**.

[0077] In one or more embodiments, the closed female luer is selected from needleless connectors/needle-free connectors. In one or more embodiments, the needleless connector is selected from a Q-Syte connector, MaxPlus, MaxPlus Clear, MaxZero, UltraSite, Caresite, In Vision-Plus, Safeline, OneLink, V-Link, ClearLink, NeutraClear, Clave, MicroClave, MicroClave Clear, Neutron, NanoClave, Kendall, Nexus, In Vision, Vadsite, Bionector, etc.
[0078] Referring to FIG. **3**, in one or more embodiments, the chamber **112** comprises a first portion **132** adjacent to the closed end **106** having a first portion diameter D**1** and a second portion **134** adjacent the open end **110** having a second portion diameter D**2**, the first portion **132** and second portion **134** in fluid communication with each other, and the first portion diameter D**1** being less than the second portion diameter D**2**.
[0079] In one or more embodiments, the annular wall **108** of the cap **102** is frusto-conically shaped.
[0080] Cap **102** is made from any of a number of types of plastic materials such as polycarbonate, polypropylene, polyethylene, glycol-modified polyethylene terephthalate, acrylonitrile butadiene styrene or any other moldable plastic material used in medical devices. In one or more embodiments, the cap comprises a polypropylene or polyethylene material. In one or more embodiments, the exterior cap surface includes a plurality of grip members.
[0081] The assembled device **200** according to a second embodiment is shown in FIGS. **8- 10**. FIGS. **11-14** show the device **200** of the second embodiment engaged with medical connectors according to embodiments of the present disclosure. Referring to FIGS. **8-10**, device **200** for connection to a medical connector according to an exemplary embodiment of the present disclosure generally comprises a cap **202**, absorbent material **214**, a disinfectant or an antimicrobial agent, and a peelable seal **228**. The cap **202** comprises an integral body **204**, a closed end **206**, an annular wall **208** having a length L.sub.C extending from the closed end **206** to an open end **210** that defines a chamber **212** containing an absorbent material **214** and disinfectant or antimicrobial agent. The open end **210** defines an end face **216** and an engagement surface **218**. In one or more embodiments, the length L.sub.C of cap **202** is shorter than/less than the length L.sub.B of the cap **102**.
[0082] Referring to FIG. **10**, the annular wall **208** of the cap **202** comprises an exterior wall surface **220** and an interior wall surface **222**. The interior wall surface **222** comprises a plurality of inwardly extending protrusions **224** sized and shaped to frictionally engage a male luer connector. In one or more embodiments, a female luer connector having a very large diameter and short length, such as a Q-Syte® in particular can also engage with the protrusion through interference/frictional fit.
[0083] The interior wall surface **222** also comprises internal threads **226** adjacent the closed end **206**. The internal threads **226** are adapted and sized to engage a female luer connector. The interior wall surface **222** defines an opening **250** adjacent the open end **210**.
[0084] In one or more embodiments, the plurality of inwardly extending protrusions **224** is sized to define an opening **130** having a diameter that is dilatable. The dilatable opening **130** can be sized to frictionally engage a male luer connector. In one or more embodiments, the dilatable opening **130** has a diameter that is dilatable from an initial diameter of from about 7-9 mm to a dilated diameter of about 9-12 mm to accept a collar of a male connector. In one or more embodiments, a female luer connector having a very large diameter and short length, such as a Q-Syte® in particular can also engage with the protrusion through interference/frictional fit.
[0085] The peelable seal **228** is disposed on the end face **216** of the cap **202** to prevent the disinfectant or the antimicrobial agent from exiting the chamber **212**.
[0086] Referring to FIGS. **8** and **9**, in one or more embodiments, the exterior wall surface **222** of the cap **202** comprises a plurality of grip members **262**.
[0087] The cap **202** comprises an annular cap wall **208** having a cap wall length Lc extending from a cap closed end **206** to a cap open end **210**. In one or more embodiments, the cap wall length Lc of cap **202** is shorter than/less than the cap wall length LB of cap **102**. The cap **202** comprises an

interior cap surface **264** and an exterior cap surface **266**.
[0088] The peelable seal **228** is disposed on the cap open end **210** to prevent the disinfectant or the antimicrobial agent from exiting the chamber.
[0089] Referring to FIG. **11**, in one or more embodiments, the male luer connector **240** frictionally engages the inwardly extending protrusions **224** via a press-fit connection upon insertion into the chamber **212**. To limit inadvertent movement of the male luer connector **240**, the plurality of inwardly extending protrusions **224** interferingly engage the male luer connector **240** in limiting movement thereof. With sufficient force being applied to the plurality of inwardly extending protrusions **224** by a male luer connector **240** when the male luer connector **240** is inserted into the open end **210** of the cap **202**, the plurality of inwardly extending protrusions **224** may be caused to deform or be displaced around the male luer connector **240** so as to nest the male luer connector **240** and thereby restrict movement of the male luer connector **240**. With sufficient force being applied to the plurality of inwardly extending protrusions **224** by a male luer connector **240** when the male luer connector **240** is inserted into the open end **210** of the cap **202**, the plurality of inwardly extending protrusions **224** may be deformed to envelope the male luer connector **240** so as to nest the male luer connector **240** and thereby restrict movement of the male luer connector **240**. In one or more embodiments, the plurality of inwardly extending protrusions **224** comprises 1-100 inwardly extending protrusions. In a specific embodiment, the plurality of inwardly extending protrusions **224** comprises 1-20 inwardly extending protrusions. In one or more embodiments, the plurality of inwardly extending protrusions **224** extends along an entire length of the interior wall surface **222** of the cap **202**. In yet another embodiment, the plurality of inwardly extending protrusions **224** partially extends along the length of the interior wall surface **222** of the cap **202**. In one or more embodiments, a female luer connector having a very large diameter and short length, such as a Q-Syte® in particular can also engage with the protrusion through interference/frictional fit.
[0090] In one or more embodiments, the plurality of inwardly extending protrusions **224** is elongated. In one or more embodiments, the plurality of inwardly extending protrusions **224** is tapered.
[0091] In one or more embodiments, the cap **202** further includes a peripheral ledge **260**
[0092] extending radially inward from the annular wall **208** which the male connector contacts when the male luer connector is inserted into the chamber **212**.
[0093] In one or more embodiments, the opening adjacent the open end **210** of the interior wall surface **222** is sized and adapted to receive a male luer connector in a press-fit connection.
[0094] Referring to FIGS. **11-13**, the interior wall surface **222** comprises internal threads **226** adjacent to the closed end **206**. The internal threads **226** are adapted and sized to engage a female luer connector **242**. In one or more embodiments, the internal threads **226** adjacent the closed end **206** of the cap **202** partially extend along a length of the interior wall surface **222** of the cap **202**.
[0095] The absorbent material **214** and the disinfectant or the antimicrobial agent contacts the male luer connector **240**, the female luer connector **242**, and the hemodialysis connector **244** after insertion of the connector through the dilatable opening **230**.
[0096] Referring to FIGS. **12** and **13**, the interior wall surface **222** comprises internal threads **226** adjacent to the closed end **206**. The internal threads **226** are adapted and sized to engage a female luer connector **242**. The absorbent material **214** and the disinfectant or the antimicrobial agent contacts the female luer connector **242** after insertion of the connector into the open end **210** of the cap **202**.
[0097] In one or more embodiments, the female connector **142** may be selected from the group consisting essentially of needle-free connectors, catheter luer connectors, stopcocks, and hemodialysis connectors.
[0098] In one or more embodiments, the male connector **240** may be an intravenous tubing end, stopcock or male lock luer.

[0099] Referring to FIG. **11**, in one or more embodiments, the male luer connector **240** rests on the peripheral ledge **260** upon being fully inserted in into the chamber **212**.
[0100] In one or more embodiments, the internal threads **226** adjacent the closed end **206** of the cap **202** partially extend along a length of the interior wall surface **222** of the cap.
[0101] In one or more embodiments, the male luer connector **240** frictionally engages the interior wall surface **222** via a press-fit connection upon insertion into the chamber **212**.
[0102] In one or more embodiments, the opening **230** adjacent the open end **210** of the interior wall surface **222** of the cap is sized and adapted to receive a male luer connector **240** in a press-fit connection.
[0103] Referring to FIG. **14**, in one or more embodiments, the interior wall surface **222** comprises internal threads **226** adjacent to the closed end **206**. The internal threads **226** are adapted and sized to engage a female connector or a needleless connector **244**. The absorbent material **214** and the disinfectant or the antimicrobial agent contacts the female connector or needleless connector **244** after insertion of the connector into the open end **210** of the cap **202**.
[0104] In one or more embodiments, the closed female luer is selected from needleless connectors/needle-free connectors. In one or more embodiments, the needleless connector **224** is selected from a Q-Syte connector, MaxPlus, MaxPlus Clear, MaxZero, UltraSite, Caresite, In Vision-Plus, Safeline, OneLink, V-Link, ClearLink, NeutraClear, Clave, MicroClave, MicroClave Clear, Neutron, NanoClave, Kendall, Nexus, In Vision, Vadsite, Bionector, etc.
[0105] Referring to FIG. **10**, in one or more embodiments, the chamber **212** comprises a first portion **232** adjacent the closed end **206** having a first portion diameter D**3** and a second portion **234** adjacent the open end **210** having a second portion diameter D**4**, the first portion **232** and second portion **234** in fluid communication with each other and the first portion diameter D**3** being less than the second portion diameter D**4**.
[0106] In one or more embodiments, the annular wall **208** of the cap **202** is frusto-conically shaped.
[0107] Cap **202** is made from any of a number of types of plastic materials such as polycarbonate, polypropylene, polyethylene, glycol-modified polyethylene terephthalate, acrylonitrile butadiene styrene or any other moldable plastic material used in medical devices. In one or more embodiments, the cap **202** comprises a polypropylene or polyethylene material. In one or more embodiments, the exterior cap surface **266** includes a plurality of grip members **262**.
[0108] Referring to FIGS. **1-7**, in one or more embodiments, the absorbent material **114** is under radial compression by the internal threads **126** to retain the absorbent material **114** in the chamber. In one or more embodiments, the absorbent material is retained in the chamber without radial compression by the internal threads. In one or more embodiments, the absorbent material **114** is a foam or a sponge. In a specific embodiment, the foam is a polyurethane foam. In a specific embodiment the absorbent material **114** is in the form of a foam plug.
[0109] The device **100** can achieve disinfection when used on luer connectors by integrating disinfectant or antimicrobial agent in the chamber **112** of the cap **102**. The disinfectant or antimicrobial agent can be directly included in the chamber **112** or disinfectant or antimicrobial agent can be absorbed into sponges or foam material that fills the chamber **112** of the cap **102**. The device **100** is designed to be compatible in interacting with various disinfectants. In one or more embodiments, the disinfectant or antimicrobial agent may include variations of alcohol or chlorhexidine. In one or more embodiments, the disinfectant or antimicrobial agent is selected from the group consisting of isopropyl alcohol, ethanol, 2-propanol, butanol, methylparaben, ethylparaben, propylparaben, propyl gallate, butylated hydroxyanisole (BHA), butylated hydroxytoluene, t-butyl-hydroquinone, chloroxylenol, chlorohexidine, chlorhexidine diacetate, chlorohexidine gluconate, povidone iodine, alcohol, dichlorobenzyl alcohol, dehydroacetic acid, hexetidine, triclosan, hydrogen peroxide, colloidal silver, benzethonium chloride, benzalkonium chloride, octenidine, antibiotic, and mixtures thereof. In a specific embodiment, the disinfectant or antimicrobial agent comprises at least one of chlorhexidine gluconate and chlorhexidine diacetate.

In one or more embodiments, the disinfectant or antimicrobial agent is a fluid or a gel.

[0110] Compression of the absorbent material **114** toward the closed end **106** of the chamber **112** upon connection to the female luer connector **142** or the male luer connector **140** or the needleless connector **144** allows the connector to contact the disinfectant or antimicrobial agent to disinfect the female luer connector **142** or the male luer connector **140** or the needleless connector **144**.

[0111] The peelable seal **128** on the end face **116** to prevent the disinfectant or the antimicrobial agent from exiting the chamber. In one or embodiments, the peelable seal **128** may be placed on the end face **116** to prevent the disinfectant or the antimicrobial agent from exiting the chamber **112**.

[0112] In one or more embodiments, the peelable seal **128** comprises an aluminum or multi-layer polymer film peel back top. In a specific embodiment, the peelable seal **128** is heat-sealed or induction sealed to the engagement surface **118** on the cap open end **110**. In one or more embodiments, the peelable seal **128** comprises a moisture barrier.

[0113] In one or more embodiments, the cap exterior wall surface **166** includes a plurality of grip members **162**.

[0114] Disinfecting caps currently on the market are capable of only disinfecting one of the three types of luer fitting, namely female luer of needle-free connectors, female luer of stopcocks, and male luer connectors on intravenous injection sites. Thus, to avoid having to use different types of disinfecting caps to clean different types of connectors, cap **102** engages with male luer connectors **140** and also with female luer connectors **142**, hemodialysis connector, and with needleless connectors **144**, thereby allowing the user to clean different types of connectors with a single device. Upon mounting the cap **102** onto female luer connectors **142**, the female luer connector **142** is inserted into the chamber **112** and screwed onto the internal threads **126** of the cap. Upon mounting the cap **102** onto a male luer connector **140**, the male luer connector **140** frictionally engages the inwardly extending protrusions **124** on the interior wall surface **122** upon insertion into the chamber **112**. In one or more embodiments, a female luer connector having a very large diameter and short length, such as a Q-Syte® in particular can also engage with the protrusion through interference/frictional fit. Hence, the device **100** of the present disclosure can be mounted onto both male and female luer connectors thus fulfilling a current need in the art.

[0115] Referring to FIGS. **8-14**, in one or more embodiments, the absorbent material **214** is under radial compression by the internal threads **226** to retain the absorbent material **214** in the chamber. In one or more embodiments, the absorbent material is retained in the chamber without radial compression by the internal threads. In one or more embodiments, the absorbent material **214** is a foam or a sponge. In a specific embodiment, the foam is a polyurethane foam. In a specific embodiment the absorbent material **214** is in the form of a foam plug.

[0116] The device **200** can achieve disinfection when used on luer connectors by integrating disinfectant or antimicrobial agent in the chamber **212** of the cap **202**. The disinfectant or antimicrobial agent can be directly included in the chamber **212** or disinfectant or antimicrobial agent can be absorbed into sponges or foam material that fills the chamber **212** of the cap **202**. The device **200** is designed to be compatible in interacting with various disinfectants. In one or more embodiments, the disinfectant or antimicrobial agent may include variations of alcohol or chlorhexidine. In one or more embodiments, the disinfectant or antimicrobial agent is selected from the group consisting of isopropyl alcohol, ethanol, 2-propanol, butanol, methylparaben, ethylparaben, propylparaben, propyl gallate, butylated hydroxyanisole (BHA), butylated hydroxytoluene, t-butyl-hydroquinone, chloroxylenol, chlorohexidine, chlorhexidine diacetate, chlorohexidine gluconate, povidone iodine, alcohol, dichlorobenzyl alcohol, dehydroacetic acid, hexetidine, triclosan, hydrogen peroxide, colloidal silver, benzethonium chloride, benzalkonium chloride, octenidine, antibiotic, and mixtures thereof. In a specific embodiment, the disinfectant or antimicrobial agent comprises at least one of chlorhexidine gluconate and chlorhexidine diacetate. In one or more embodiments, the disinfectant or antimicrobial agent is a fluid or a gel.

[0117] Compression of the absorbent material **214** toward the closed end **106** of the chamber **212** upon connection to the female luer connector **242** or the male luer connector **140** or the needleless connector **244** allows the connector to contact the disinfectant or antimicrobial agent to disinfect the female luer connector **242** or the male luer connector **240** or the needleless connector **244**.

[0118] The peelable seal **228** on the end face **216** to prevent the disinfectant or the antimicrobial agent from exiting the chamber. In one or embodiments, the peelable seal **228** may be placed on the end face **216** to prevent the disinfectant or the antimicrobial agent from exiting the chamber **212**.

[0119] In one or more embodiments, the peelable seal **228** comprises an aluminum or multi-layer polymer film peel back top. In a specific embodiment, the peelable seal **228** is heat-sealed or induction sealed to the engagement surface **218** on the cap open end **210**. In one or more embodiments, the peelable seal **228** comprises a moisture barrier.

[0120] In one or more embodiments, the cap exterior wall surface **266** includes a plurality of grip members **262**.

[0121] Disinfecting caps currently on the market are capable of only disinfecting one of the three types of luer fitting, namely female luer of needle-free connectors, female luer of stopcocks, and male luer connectors on intravenous injection sites. Thus, to avoid having to use different types of disinfecting caps to clean different types of connectors, cap **202** engages with male luer connectors **240** and also with female luer connectors **242** and with needleless connectors **244**, thereby allowing the user to clean different types of connectors with a single device. Upon mounting the cap **202** onto female luer connectors **242**, the female luer connector **242** is inserted into the chamber **212** and screwed onto the internal threads **226** of the cap. Upon mounting the cap **202** onto a male luer connector **240**, the male luer connector **240** frictionally engages the inwardly extending protrusions **224** on the interior wall surface **222** upon insertion into the chamber **212**. Hence, the device **200** of the present disclosure can be mounted onto both male and female luer connectors thus fulfilling a current need in the art.

[0122] Referring to FIGS. **15** to **18**, in one or more embodiments, the cap of the device of the present disclosure may form a fluid-tight seal with a female luer connector **300**, a male luer connector **400** or hemodialysis connector **500**. Referring to FIGS. **15** to **18**, in one or more embodiments, the cap of the device of the present disclosure is tapered to form a fluid-tight seal with a male luer connector **400**. In specific embodiments, the cap is compliant with ISO standards (e.g., ISO 594-1:1986 and ISO 594-2:1998) for forming a seal with a male luer connector.

[0123] In one or more embodiments, the cap of the device of the present disclosure has threads that have a size and pitch to engage a threadable segment of a female connector, such as for example, a female luer connector. Such connectors are generally and commonly used as catheter and other fluid-tight protective connectors in medical applications. In some embodiments, the cap provides a protective cover for a female luer connector when engaged with the connector when threads from the female luer connector engage and form a releasable connection with threads of the cap.

[0124] In some embodiments, the connector comprises a needleless injection site, which may sometimes be referred to as a needleless injection port, hub, valve, or device, or as a needleless access site, port, hub, valve, or device, and which can include such brands as, for example, Clave® (available from ICU Medical, Inc.), SmartSite® (available from Cardinal Health, Inc.), and Q-Syte™ (available from Becton, Dickinson and Company). In some embodiments, the cap can be connected with any of a variety of different needleless injection sites, such as those previously listed. In one or more embodiments, after the cap has been coupled with connector, it is unnecessary to disinfect (e.g. treat with an alcohol swab) the connector prior to each reconnection of the connector with another connector, as the connector will be kept in an uncontaminated state while coupled with the cap. Use of the cap replaces the standard swabbing protocol for cleaning connectors.

[0125] In one or more embodiments, threads of the cap are sized and pitched to engage threads of a male luer-lock connector. For example, connector can comprise the end of an IV tubing set that is disconnected from an IV catheter needleless injection site.

[0126] Other aspects of the present disclosure are directed to methods of disinfecting medical connectors and assemblies. In one or more embodiments, a method of disinfecting a medical connector comprises connecting the device of one or more embodiments to a medical connector, wherein connecting includes frictionally engaging the interior wall surface upon insertion into the chamber or engaging the internal threads, such that the medical connector contacts the absorbent material and the disinfectant or antimicrobial agent.
[0127] In one or more embodiments, an assembly comprises the device of one or more embodiments connected to a medical connector. In one or more embodiments, the medical connector is selected from a male luer connector, a female luer connector, and a needleless connector.
[0128] Reference throughout this specification to “one embodiment,” “certain embodiments,” “one or more embodiments” or “an embodiment” means that a particular feature, structure, material, or characteristic described in connection with the embodiment is included in at least one embodiment of the disclosure. Thus, the appearances of the phrases such as “in one or more embodiments,” “in certain embodiments,” “in one embodiment” or “in an embodiment” in various places throughout this specification are not necessarily referring to the same embodiment of the disclosure. Furthermore, the particular features, structures, materials, or characteristics may be combined in any suitable manner in one or more embodiments.
[0129] Although the disclosure herein has provided a description with reference to particular embodiments, it is to be understood that these embodiments are merely illustrative of the principles and applications of the present disclosure. It will be apparent to those skilled in the art that various modifications and variations can be made to the method and apparatus of the present disclosure without departing from the spirit and scope of the disclosure. Thus, it is intended that the present disclosure include modifications and variations that are within the scope of the appended claims and their equivalents.

Claims

1. A method of disinfecting a medical connector, the method comprising: connecting a device having a cap comprising an integral body, a closed end, an annular wall having a length LB extending from the closed end to an open end and defining a chamber containing an absorbent material and disinfectant or antimicrobial agent, the open end defining an end face, the open end defining an engagement surface; the annular wall having an exterior wall surface and an interior wall surface comprising a plurality of inwardly extending protrusions sized and shaped to frictionally engage a male luer connector or a female luer connector; the interior wall surface comprising internal threads adjacent the closed end, the internal threads adapted and sized to engage a female luer connector, wherein connecting comprises frictionally engaging the interior wall surface upon insertion into the chamber such that the medical connector contacts the absorbent material and the disinfectant or antimicrobial agent.
2. The method of claim 1, wherein the female luer connector is selected from the group consisting of needle-free connectors, catheter luer connectors, stopcocks, and hemodialysis connectors.
3. The method of claim 1, wherein the male luer connector is an intravenous tubing end, stopcock, or male lock luer.
4. The method of claim 3, wherein the step of connecting further comprising mounting the cap onto the male luer connector by frictionally engaging the male luer connector to the inwardly extending protrusions on the interior wall surface via a press-fit connection upon insertion into the chamber.
5. The method of claim 4, wherein insertion of a collar of a male connector dilates the opening defined by the plurality of inwardly extending protrusions from an initial diameter of from about 7-9 mm to a dilated diameter of about 9-12 mm to accept the male connector.
6. The method of claim 5, further comprising applying force to the plurality of inwardly extending

protrusions by inserting the male luer connector into the open end of the cap to deform or displace the plurality of inwardly extending protrusions around the male luer connector to nest the male luer connector.

7. The method of claim 6, further comprising inserting the male luer connector into the chamber to contact and rest on the peripheral ledge upon being fully inserted in into the chamber.

8. The method of claim 7, further comprising compressing the absorbent material toward the closed end of the chamber upon connection to the male luer connector to release the disinfectant or antimicrobial agent to contact and disinfect the male luer connector.

9. The method of claim 2, wherein the step of connecting further comprising mounting the cap onto the female luer connector by inserting the female luer connector into the opening of the chamber to contact the internal threads.

10. The method of claim 6, further comprising screwing the female luer connector onto the internal threads to engage the internal threads.

11. The method of claim 7, further comprising screwing the female luer connector onto the internal threads creating a fluid tight seal.

12. The method of claim 8, further comprising compressing the absorbent material toward the closed end of the chamber upon connection to the female luer connector to release the disinfectant or antimicrobial agent to contact and disinfect the female luer connector.